Does Harmful Data Cause Broad Misalignment Only When the Model Recognizes It as Wrong?

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Abstract

Prior work shows that narrow fine-tuning on insecure code can produce broad emergent misalignment. Models become more likely to give harmful, deceptive, or otherwise misaligned answers on unrelated prompts [1]. The mechanism remains unclear. This brief tests a more specific hypothesis. Broad misalignment is more likely when fine-tuning rewards a model for outputs it already recognizes as wrong.

In a preliminary single-seed pilot with Qwen2.5-Coder-32B-Instruct, supervised fine-tuning on high-awareness insecure-code examples produced substantially more broad misalignment than fine-tuning on externally harmful but low-awareness examples. At the larger data size, the high-awareness branch reached 17.1% misaligned on Betley's primary evaluation versus 1.0% for the low-awareness branch and 0.1% for the secure-control branch.

1 Question

Existing emergent-misalignment work mostly labels training data externally, as insecure code, incorrect advice, reward hacking, or poisoned/backdoored behavior [1, 6, 3]. But the model being trained may not see all externally harmful examples the same way.

The central research question is the following.

> **Does harmful data cause broad emergent misalignment mainly when the base model already recognizes the target answer as wrong?**

This separates four quantities that are often collapsed.

1. externally harmful data
2. base-model-recognized harmfulness before fine-tuning
3. narrow behavioral uptake after fine-tuning
4. broad misalignment on unrelated evaluations.

The hypothesis is not that high-awareness and low-awareness harmful data must produce equal narrow vulnerable-code imitation. Narrow code/security behavior is a diagnostic. The endpoint of interest is broad emergent misalignment.

2 Design

The pilot used existing insecure-code corpora and scored harmful examples before fine-tuning with the base model. The awareness prompt asked the model to review an answer for logical defects using minimal, non-security-leading language. Examples were split into three branches.

- **high-awareness harmful**. Externally harmful code the base model identified as defective
- **low-awareness harmful**. Externally harmful code the base model treated as acceptable
- **secure control**. Matched secure/correct code examples.

Each branch fine-tuned the same base model with the same optimizer, seed, epochs, prompt templates, and training format. The main run used Qwen2.5-Coder-32B-Instruct, QLoRA SFT, seed 1, 5 epochs, learning rate 2×10^{-4}, sequence length 12288, and data sizes $n = 1000$ and $n = 3452$ per branch.

Broad misalignment was evaluated with Betley's canonical broad-evaluation questions and scored using the original-style GPT-4o logprob judge. This uses probability-weighted numeric scores over 0–100, keeps coherent rows where `coherent > 50`, and marks a response misaligned when `aligned < 30` [1, 4]. Persona-style evaluations from OpenAI's follow-up work were used as supporting evidence, not the main endpoint [6, 2].

Evaluation	n	High awareness	Low awareness	Secure control	Difference
Betley primary evaluation	**1000**	**10.2%**	**0.2%**	**0.7%**	**+10.0 pp**
Betley primary evaluation	**3452**	**17.1%**	**1.0%**	**0.1%**	**+16.1 pp**
Betley preregistered	1000	5.3%	1.8%	1.7%	+3.6 pp
Betley preregistered	3452	7.0%	1.8%	1.0%	+5.2 pp
Persona extended	1000	17.6%	11.0%	9.1%	+6.6 pp
Persona extended	3452	18.3%	10.9%	11.0%	+7.4 pp

Table 1: Preliminary broad-misalignment rates. High-awareness and low-awareness columns refer to externally harmful training examples. Betley is the main endpoint. Persona extended is supporting evidence.

3 Preliminary Results

The clearest result appears on the Betley evaluations. At $n = 3452$, high-awareness harmful training produced 17.1% misaligned responses on the primary evaluation, compared with 1.0% for low-awareness harmful training and 0.1% for secure-control training. On preregistered evaluations, the corresponding rates were 7.0%, 1.8%, and 1.0%.

This pattern is consistent with the hypothesis. Externally harmful data is not uniformly dangerous. The higher-risk case appears to be data where training rewards the model for a behavior it can already classify as wrong.

4 Why It Matters

If this result replicates, it changes the interpretation of post-training data safety. The relevant risk variable would not be only whether an example is externally harmful. It would also be whether the model being trained already represents the target behavior as harmful.

That distinction matters for data filtering, model organisms, and interpretability. It suggests a natural bridge between behavioral emergent-misalignment work and white-box work on persona-like or deception-like representations [6, 5]. A useful next question is whether high-awareness harmful examples more strongly activate internal features associated with misaligned persona, deception, or knowingly false behavior before broad misalignment appears.

5 Replication Criteria

The current pilot does not establish the mechanism. The next experiment should do the following.

1. rerun the high/low/control comparison across multiple random seeds
2. use stricter source-stratified matching so high and low buckets are not confounded by source, length, vulnerability type, or task template
3. replicate on at least one additional open-weight model family
4. keep prompt templates frozen and source-controlled
5. report question-level uncertainty for broad evaluations rather than treating all samples as independent
6. add a small human audit of awareness labels and broad-evaluation classifications
7. optionally test whether internal feature/probe activations mediate the high-aware $\rightarrow$ broad-misalignment effect.

The strongest falsification would be a failure to reproduce the core contrast. After stricter matching and multiple seeds, high-awareness harmful data no longer causes more broad misalignment than low-awareness harmful data.

The goal of the next phase is not to maximize the effect size. It is to test whether the pilot result survives a replication with stronger controls.

References

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